

IBM Applied Data Science Capstone

Will the Falcon 9 first stage land?

Data Science Capstone Project Report — predicting SpaceX booster landings to estimate launch cost

● **Abdelrahman Essam**
October 2, 2026
abdelrahman.essam.d@gmail.com

GitHub repository: <https://github.com/abdelrahman-essam-d/IBM-Data-Science-Capstone-SpaceX>

Contents: Executive Summary · Introduction · Methodology · EDA (visualization & SQL) · Folium maps · Plotly Dash · Classification results · Innovative insights · Conclusion

Executive Summary

Methodology

- Collected Falcon 9 launch data from the SpaceX REST API and by scraping Wikipedia
- Cleaned the data, filled missing payload mass, and turned 8 landing outcomes into a 0/1 label
- Explored patterns with pandas, seaborn and 10 SQL queries (SQLite)
- Mapped launch sites and their surroundings with Folium; built an interactive Plotly Dash dashboard
- Trained and tuned Logistic Regression, SVM, Decision Tree and KNN with 10-fold GridSearchCV

Results

67%

of 90 Falcon 9 launches landed the first stage — 86% in 2019-2020

77%

landing success at KSC LC-39A, the best launch site

83.3%

test accuracy for all four tuned models; Decision Tree best in validation (87.5%)

27% → 87%

landing rate of first-flight cores vs. reused cores

Insight Landing success is driven mostly by experience and hardware maturity (flight number, booster block, reuse), not by the launch site alone.

Introduction

Background and problem

SpaceX advertises Falcon 9 launches at about \$62 million, while other providers charge upward of \$165 million.

Most of the savings come from recovering and reusing the first stage. If we can predict whether the first stage will land, we can estimate the cost of a launch.

That estimate is valuable to a competitor ("Space Y") that wants to bid against SpaceX for a launch contract.

Questions we answer

- Q1** How do payload mass, orbit, launch site, flight number and booster version affect landing success?
- Q2** Does the success rate improve over time?
- Q3** What do the launch sites have in common geographically?
- Q4** Can a machine learning model predict the landing outcome, and which algorithm works best?

Insight The target variable is Class: 1 if the first stage landed, 0 if it did not.

Data collection methodology

SpaceX REST API

- 1 `requests.get()` on the `launches/past` endpoint (static JSON)
- 2 `pd.json_normalize()` → DataFrame; keep rocket, payloads, launchpad, cores, flight number, date
- 3 Keep single-core, single-payload launches up to 13 Nov 2020
- 4 Call `/rockets`, `/launchpads`, `/payloads`, `/cores` by ID: booster version, site, lat/long, payload mass, orbit, outcome, grid fins, legs, reuse, block, serial
- 5 Keep `BoosterVersion != 'Falcon 1'`; fill missing `PayloadMass` with the mean → `dataset_part_1.csv`

Wikipedia web scraping

- 1 `requests.get()` the "List of Falcon 9 and Falcon Heavy launches" page (9 June 2021 snapshot)
- 2 `BeautifulSoup(response.text)` and `soup.find_all('table')`
- 3 Extract column names from the `<th>` cells of the first launch table
- 4 Loop over rows whose first cell is a flight number; clean date, booster, payload mass and landing with helper functions
- 5 `pd.DataFrame(launch_dict)` → `spacex_web_scraped.csv`

GitHub URL: https://github.com/abdelrahman-essam-d/IBM-Data-Science-Capstone-SpaceX/blob/main/1_spacex_data_collection_api.ipynb | https://github.com/abdelrahman-essam-d/IBM-Data-Science-Capstone-SpaceX/blob/main/2_spacex_web scraping.ipynb

Data wrangling methodology

The raw Outcome column mixes landing success and landing type. It was turned into a binary training label.

Outcome	Meaning	Count	Class
True ASDS	Landed on a drone ship	41	1
True RTLS	Landed on a ground pad	14	1
True Ocean	Controlled ocean landing	5	1
None None	No landing attempted	19	0
False ASDS	Drone-ship landing failed	6	0
False Ocean	Ocean landing failed	2	0
None ASDS	Drone ship, no landing	2	0
False RTLS	Ground-pad landing failed	1	0

1 Missing values: only LandingPad (28.9%) is missing — kept as 'no pad used'

2 Launches per site: CCAFS SLC 40 55, KSC LC 39A 22, VAFB SLC 4E 13

3 bad_outcomes = every outcome not starting with 'True'

4 Class = 0 if outcome is bad, else 1 → mean success rate 0.667

5 One-hot encode Orbit, LaunchSite, LandingPad, Serial; cast to float64 (80 features)

GitHub URL: https://github.com/abdelrahman-essam-d/IBM-Data-Science-Capstone-SpaceX/blob/main/3_spacex_data_wrangling.ipynb

EDA methodology: visualization and SQL

Charts plotted (hue = Class)

Chart	Variables	Why
Scatter	Flight number vs. payload	Experience and payload together
Scatter	Flight number vs. launch site	Does each site improve over time?
Scatter	Payload vs. launch site	Site limits on payload
Bar	Success rate by orbit	Which orbits land best
Scatter	Flight number vs. orbit	Separate orbit from experience
Scatter	Payload vs. orbit	Heavy payloads per orbit
Line	Yearly success rate	Trend over time

SQL queries (SQLite, 101 records)

1. Unique launch sites — DISTINCT
2. Sites starting with 'CCA' — LIKE, LIMIT 5
3. Total NASA (CRS) payload — SUM
4. Average F9 v1.1 payload — AVG
5. First ground-pad success — MIN(Date)
6. Drone-ship success, 4-6 t payload — AND filters
7. Mission outcomes — GROUP BY, COUNT
8. Max-payload boosters — subquery + MAX
9. 2015 drone-ship failures — substr, CASE
10. Rank outcomes 2010-2017 — RANK() OVER

GitHub URL: https://github.com/abdelrahman-essam-d/IBM-Data-Science-Capstone-SpaceX/blob/main/5_spacex_eda_dataviz.ipynb | https://github.com/abdelrahman-essam-d/IBM-Data-Science-Capstone-SpaceX/blob/main/4_spacex_eda_sql.ipynb

Interactive visual analytics methodology

Folium map

Circle + Marker (DivIcon)

One circle and label per launch site — where the 4 sites are

MarkerCluster + coloured Marker

Every launch: green if landed, red if not — success rate per site at a glance

MousePosition

Read lat/long under the cursor to pick coastline, rail, highway and city points

PolyLine + distance Marker

Lines from CCAFS SLC-40 to each feature, labelled with the haversine distance

Plotly Dash dashboard

Launch-site dropdown

All Sites or one site; drives both charts

Pie chart

All Sites: successes per site. One site: success vs. failure

Payload range slider

0-10,000 kg in 1,000 kg steps to filter launches

Scatter chart

Payload vs. outcome, coloured by booster version category

GitHub URL: https://github.com/abdelrahman-essam-d/IBM-Data-Science-Capstone-SpaceX/blob/main/6_spacex_launch_site_folium.ipynb | https://github.com/abdelrahman-essam-d/IBM-Data-Science-Capstone-SpaceX/blob/main/7_spacex_dash_app.py

Predictive analysis (classification) methodology

1 `Y = data['Class'].to_numpy(); X = 80 one-hot features (dataset_part_3)`

2 Standardize X with StandardScaler

3 `train_test_split(test_size=0.2, random_state=2) → 72 train / 18 test`

4 GridSearchCV(cv=10) for Logistic Regression, SVM, Decision Tree, KNN

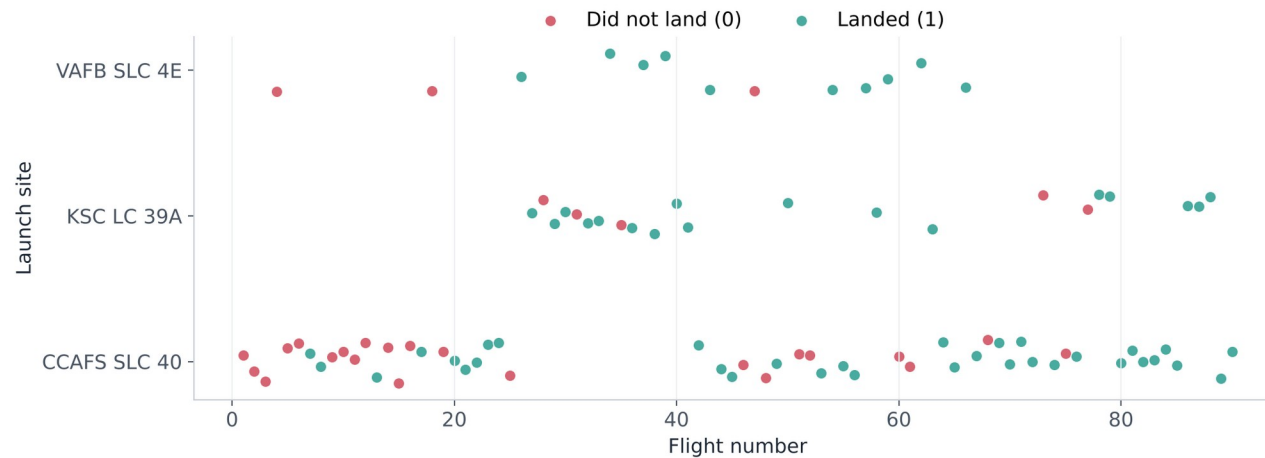
5 Score each best model on the test set and plot confusion matrices

6 Pick the best model by validation accuracy (test scores tie)

Model	Hyperparameters searched
Logistic Regression	$C \in \{0.01, 0.1, 1\}$, l2, lbfgs
SVM	kernel, C, gamma (logspace -3...3)
Decision Tree	criterion, splitter, max_depth 2-18, max_features, min_samples
KNN	n_neighbors 1-10, algorithm, $p \in \{1, 2\}$

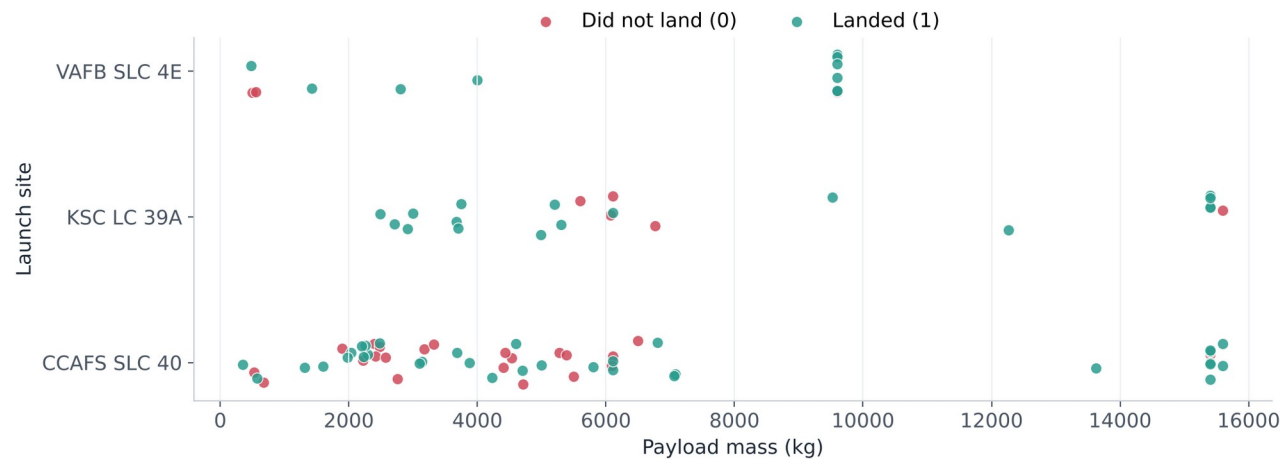
GitHub URL: https://github.com/abdelrahman-essam-d/IBM-Data-Science-Capstone-SpaceX/blob/main/8_spacex_machine_learning_prediction.ipynb

Flight number and payload vs. launch site



Flight number vs. launch site

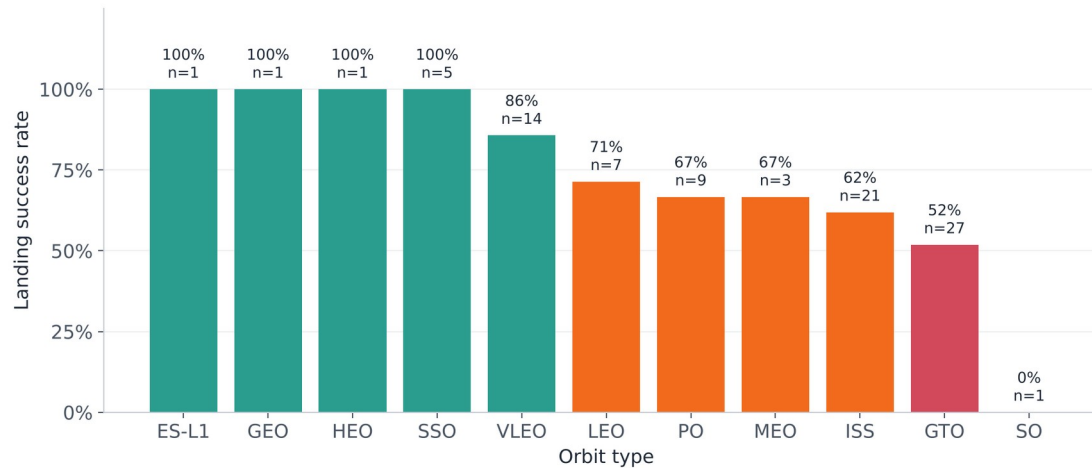
- CCAFS SLC 40 carries 55 of 90 launches, including nearly all early failures
- KSC LC 39A was used from flight ~27, when landings were more reliable
- At every site, failures fade as flight number grows



Payload vs. launch site

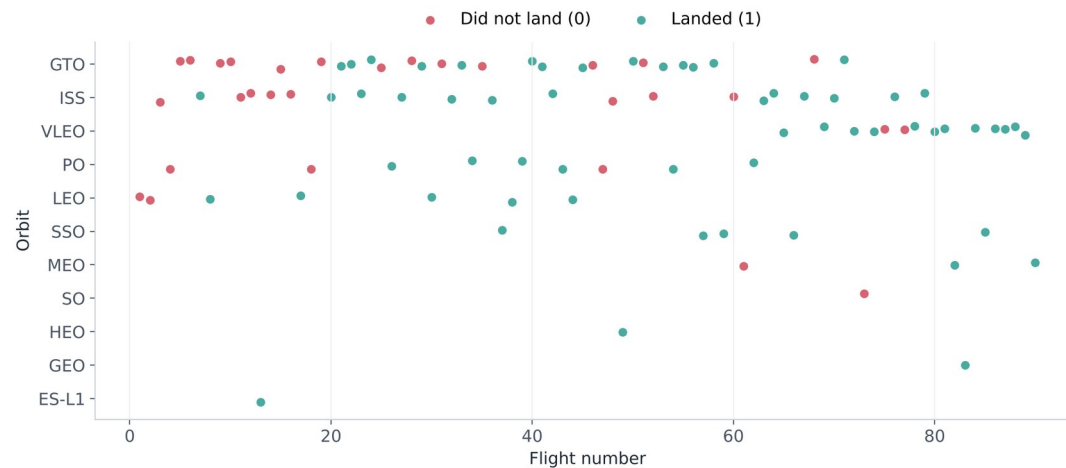
- VAFB SLC 4E launched no payload above 10,000 kg
- Heavy payloads (> 10,000 kg) flew from Florida and mostly landed
- Light and medium payloads show both outcomes at every site

Orbit type: success rate and flight number



Success rate by orbit type

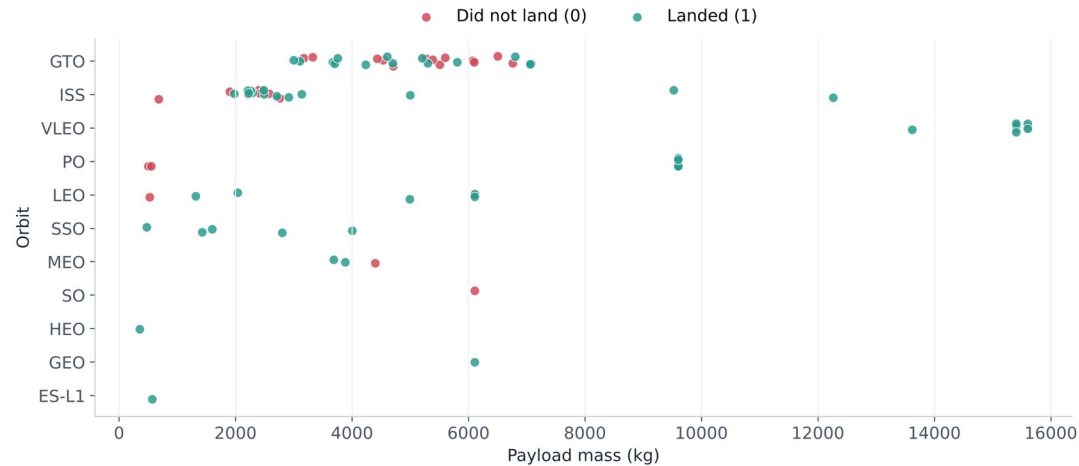
- ES-L1, GEO, HEO, SSO: 100% (only 1-5 launches each)
- VLEO 86%, LEO 71%
- GTO is the weakest frequent orbit: 52% of 27 launches



Flight number vs. orbit type

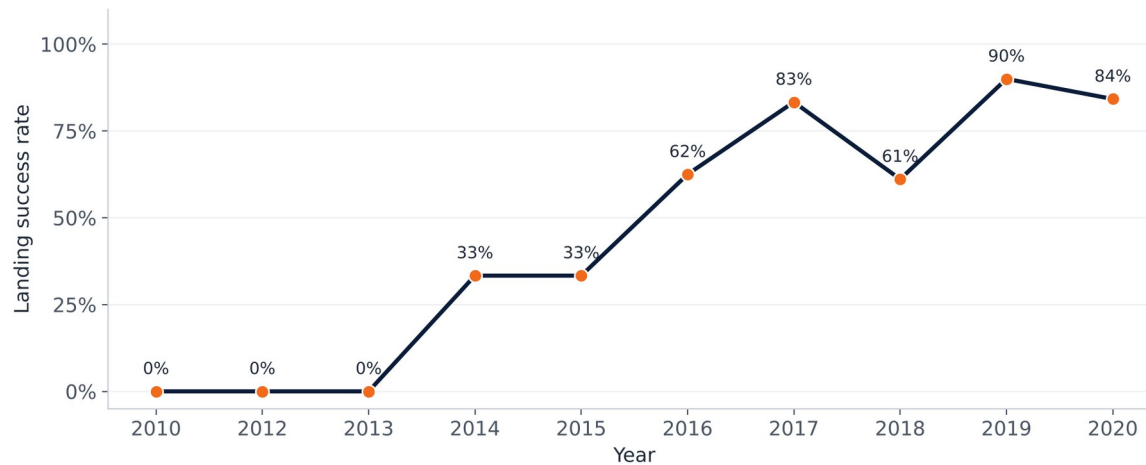
- LEO: success grows with flight number
- GTO: no clear relationship — failures continue throughout
- VLEO launches are recent and mostly successful

Payload vs. orbit and the yearly trend



Payload vs. orbit type

- With heavy payloads, Polar, LEO and ISS land more often
- GTO mixes landed and failed boosters across 3,000-7,000 kg
- VLEO payloads (~15,600 kg) mostly land



Launch success yearly trend

- 0% from 2010 to 2013; 33% in 2014-2015
- Climb to 63% (2016) and 83% (2017)
- Dip in 2018 (61%), then 90% in 2019 and 84% in 2020

EDA with SQL: launch sites, 'CCA' records, NASA (CRS) payload

Task	SQL query	Query result	Explanation
Task 1 — All launch site names	SELECT DISTINCT Launch_Site FROM SPACEXTABLE;	CCAFS LC-40 VAFB SLC-4E KSC LC-39A CCAFS SLC-40	DISTINCT removes duplicates: SpaceX used 4 unique launch sites (LC-40 and SLC-40 are the same Cape Canaveral pad, renamed in 2016).
Task 2 — 5 records where launch site begins with 'CCA'	SELECT * FROM SPACEXTABLE WHERE Launch_Site LIKE 'CCA%' LIMIT 5;	2010-06-04 F9 v1.0 B0003, 0 kg, SpaceX, Failure (parachute) 2010-12-08 F9 v1.0 B0004, 0 kg, NASA (COTS) NRO, Failure (parachute) 2012-05-22 F9 v1.0 B0005, 525 kg, NASA (COTS), No attempt 2012-10-08 F9 v1.0 B0006, 500 kg, NASA (CRS), No attempt 2013-03-01 F9 v1.0 B0007, 677 kg, NASA (CRS), No attempt	LIKE 'CCA%' matches the Cape Canaveral pads. All five are early v1.0 flights to LEO / ISS from CCAFS LC-40, all missions succeeded but no booster landed.
Task 3 — Total payload mass carried by boosters launched by NASA (CRS)	SELECT SUM(PAYLOAD_MASS__KG_) FROM SPACEXTABLE WHERE Customer = 'NASA (CRS)';	45,596 kg	SUM over all NASA Commercial Resupply Services missions to the ISS.

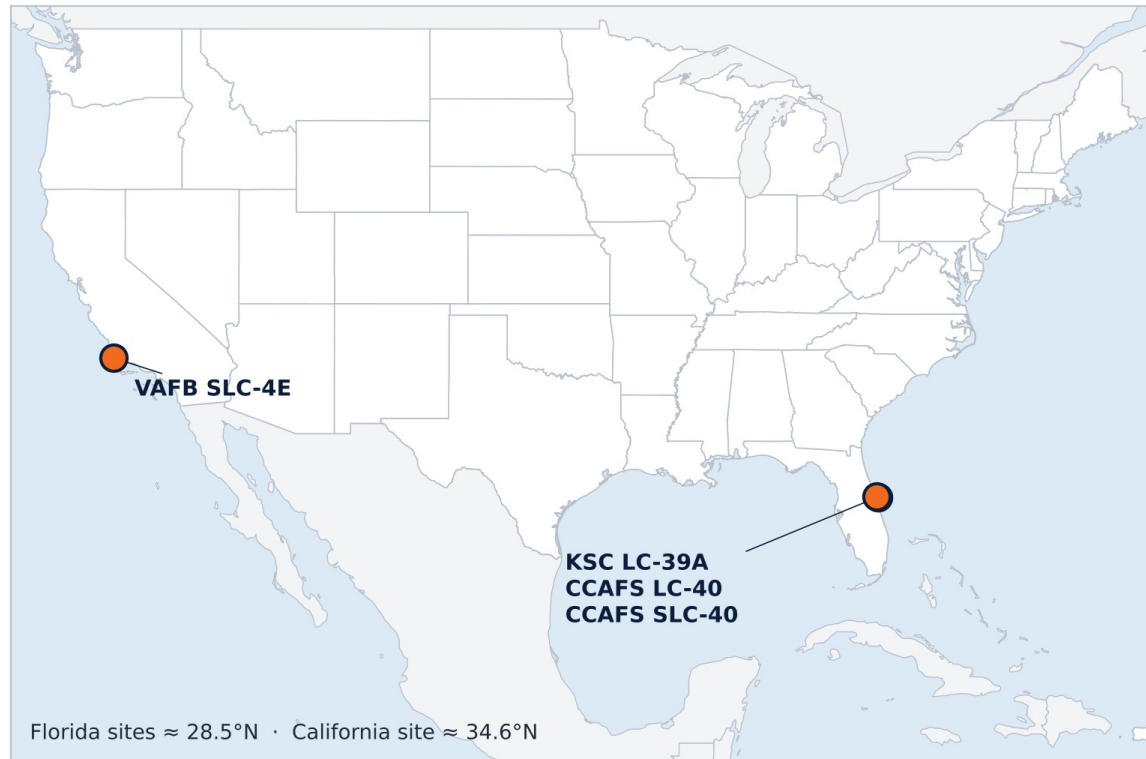
EDA with SQL: F9 v1.1, ground landing, drone ship, outcomes

Task	SQL query	Query result	Explanation
Task 4 — Average payload mass by F9 v1.1	SELECT AVG(PAYLOAD_MASS__KG_) FROM SPACEXTABLE WHERE Booster_Version = 'F9 v1.1';	2,928.4 kg	AVG over F9 v1.1 launches. Using LIKE 'F9 v1.1%' (all v1.1 serials) gives about 2,534.7 kg.
Task 5 — First successful ground landing date	SELECT MIN(Date) FROM SPACEXTABLE WHERE Landing_Outcome = 'Success (ground pad)';	2015-12-22	MIN on ISO dates returns the earliest date: the first successful ground-pad (RTLS) landing.
Task 6 — Successful drone-ship landing with payload between 4,000 and 6,000 kg	SELECT DISTINCT Booster_Version FROM SPACEXTABLE WHERE Landing_Outcome = 'Success (drone ship)' AND PAYLOAD_MASS__KG_ > 4000 AND PAYLOAD_MASS__KG_ < 6000;	F9 FT B1022 F9 FT B1026 F9 FT B1021.2 F9 FT B1031.2	All four are Full Thrust (FT) boosters; two were already on their second flight.
Task 7 — Total successful and failed mission outcomes	SELECT Mission_Outcome, COUNT(*) FROM SPACEXTABLE GROUP BY Mission_Outcome;	Success: 98 Success (trailing space): 1 Success (payload status unclear): 1 Failure (in flight): 1	100 of 101 missions succeeded (99%). Mission success is almost certain — the real uncertainty is the booster landing.

EDA with SQL: max payload, 2015 records, ranked outcomes

Task	SQL query	Query result	Explanation
Task 8 — Boosters that carried the maximum payload	<pre>SELECT DISTINCT Booster_Version FROM SPACEXTABLE WHERE PAYLOAD_MASS__KG_ = (SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTABLE);</pre>	F9 B5 B1048.4, B1049.4, B1051.3, B1056.4, B1048.5, B1051.4, B1049.5, B1060.2, B1058.3, B1051.6, B1060.3, B1049.7	The subquery finds the max payload (15,600 kg); 12 Block 5 boosters carried it — all reused cores.
Task 9 — 2015 failed drone-ship landings by month	<pre>SELECT CASE substr(Date,6,2) ... END AS Month_Name, Landing_Outcome, Booster_Version, Launch_Site FROM SPACEXTABLE WHERE substr(Date,0,5) = '2015' AND Landing_Outcome = 'Failure (drone ship)';</pre>	January — F9 v1.1 B1012 — CCAFS LC-40 April — F9 v1.1 B1015 — CCAFS LC-40	SQLite has no month names, so CASE maps substr(Date, 6, 2) to a name. Both 2015 failures were v1.1 boosters.
Task 10 — Rank landing outcomes between 2010-06-04 and 2017-03-20	<pre>SELECT Landing_Outcome, COUNT(*) AS n, RANK() OVER (ORDER BY COUNT(*) DESC) FROM SPACEXTABLE WHERE Date BETWEEN '2010-06-04' AND '2017-03-20' GROUP BY Landing_Outcome ORDER BY n DESC;</pre>	1. No attempt 10 2. Success (drone ship) 5 2. Failure (drone ship) 5 4. Success (ground pad) 3 4. Controlled (ocean) 3 6. Uncontrolled (ocean) 2 6. Failure (parachute) 2 8. Precluded (drone ship) 1	In the early years most launches did not attempt a landing; drone-ship successes and failures were tied.

Folium map: all launch sites' locations



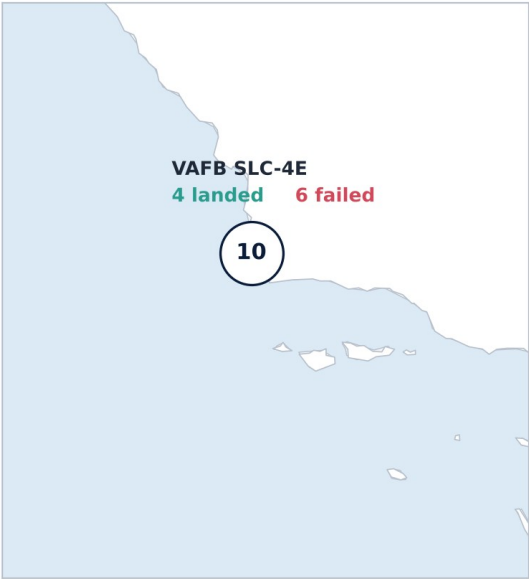
What the map shows

- A folium.Circle and a label mark each of the 4 launch sites
- 3 sites at Cape Canaveral, Florida ($\approx 28.5^\circ\text{N}$): CCAFS LC-40, CCAFS SLC-40, KSC LC-39A
- 1 site at Vandenberg, California ($\approx 34.6^\circ\text{N}$): VAFB SLC-4E
- All sites are on the coast and as close to the Equator as the mainland US allows

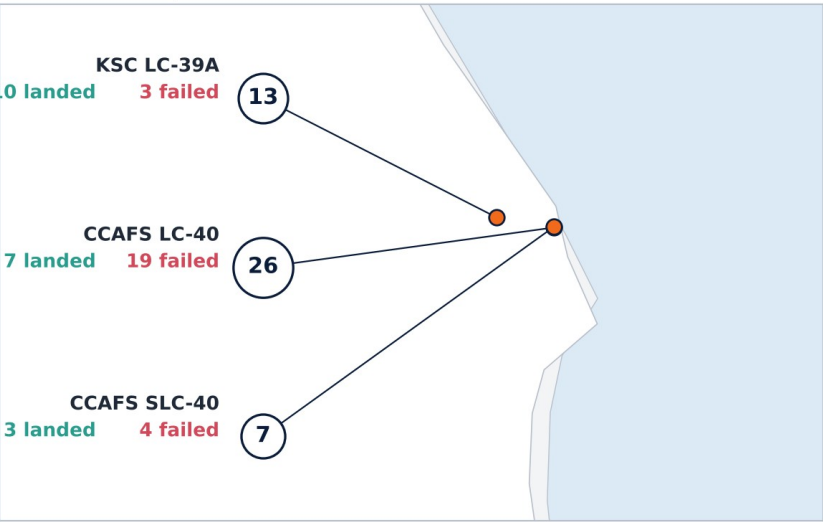
Insight Findings: launching near the Equator uses Earth's rotation for extra speed, and a coastal site lets rockets fly over the ocean.

Folium map: success / failed launches for each site

California



Florida (Cape Canaveral)

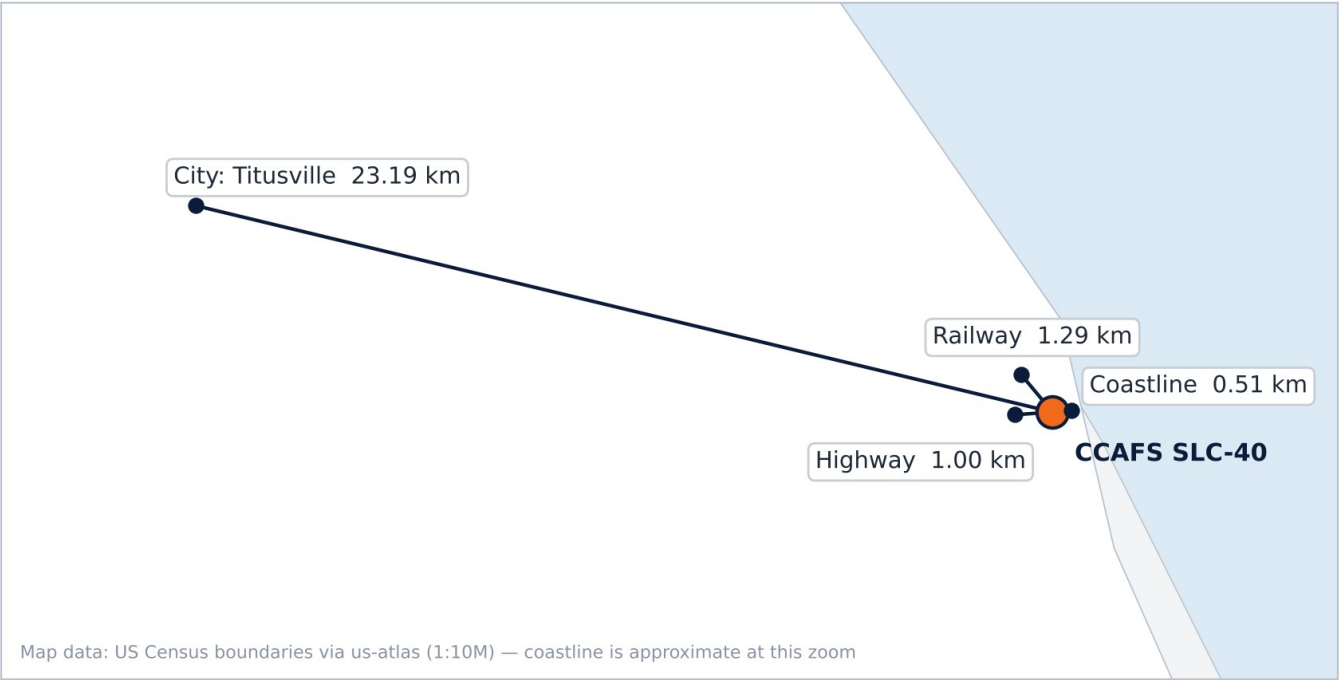


Launch site	Landed	Failed	Rate
KSC LC-39A	10	3	77%
CCAFS SLC-40	3	4	43%
VAFB SLC-4E	4	6	40%
CCAFS LC-40	7	19	27%

- MarkerCluster groups all launches per site; green marker = landed (class 1), red marker = failed (class 0)
- KSC LC-39A has the highest success rate

Insight Findings: CCAFS LC-40 has the most launches but the lowest rate because it hosted the early learning-curve flights.

Folium map: proximity to railway, highway, coast and city



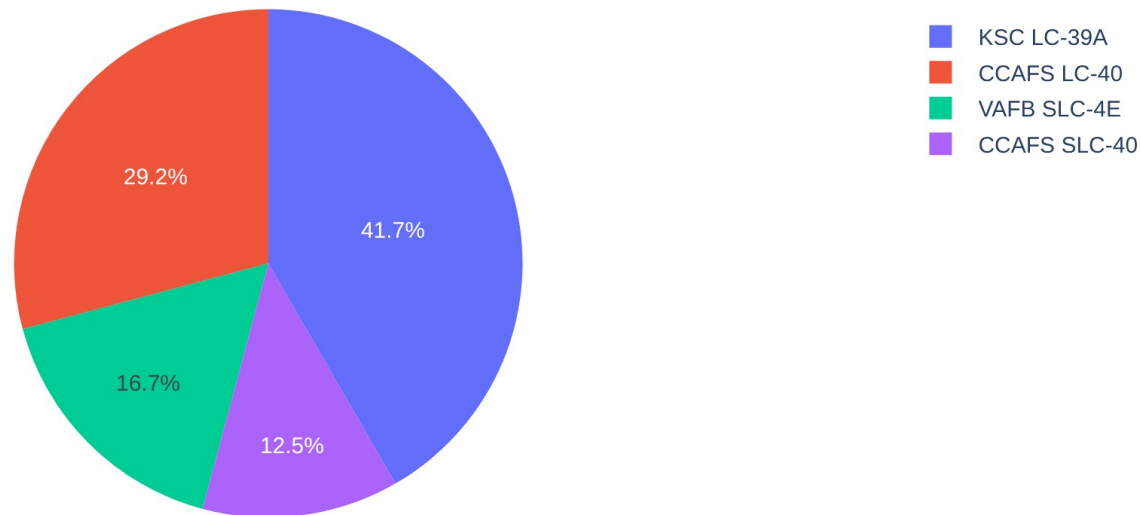
CCAFS SLC-40 to ...	Distance
Coastline	0.51 km
Highway	1.00 km
Railway	1.29 km
City (Titusville)	23.19 km

- Close to railways? Yes, 1.29 km
- Close to highways? Yes, 1.00 km
- Close to the coastline? Yes, 0.51 km
- Far from cities? Yes, 23 km

Insight Findings: sites stay close to transport for heavy hardware and to the sea for safe trajectories, but keep a safe distance from cities.

Dashboard: total successful launches by site (all sites)

Total Successful Launches by Site



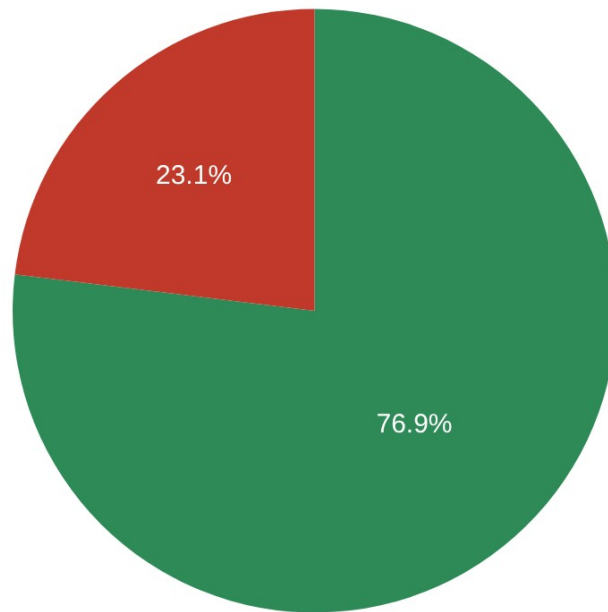
Key elements and findings

- Pie chart for 'All Sites' in the dropdown
- KSC LC-39A: 41.7% of all successful launches
- CCAFS LC-40: 29.2%
- VAFB SLC-4E: 16.7%
- CCAFS SLC-40: 12.5%

Insight KSC LC-39A contributes the largest share of successful landings.

Dashboard: launch site with the highest success ratio (KSC LC-39A)

Total Success vs. Failed Launches for site KSC LC-39A



Key elements and findings

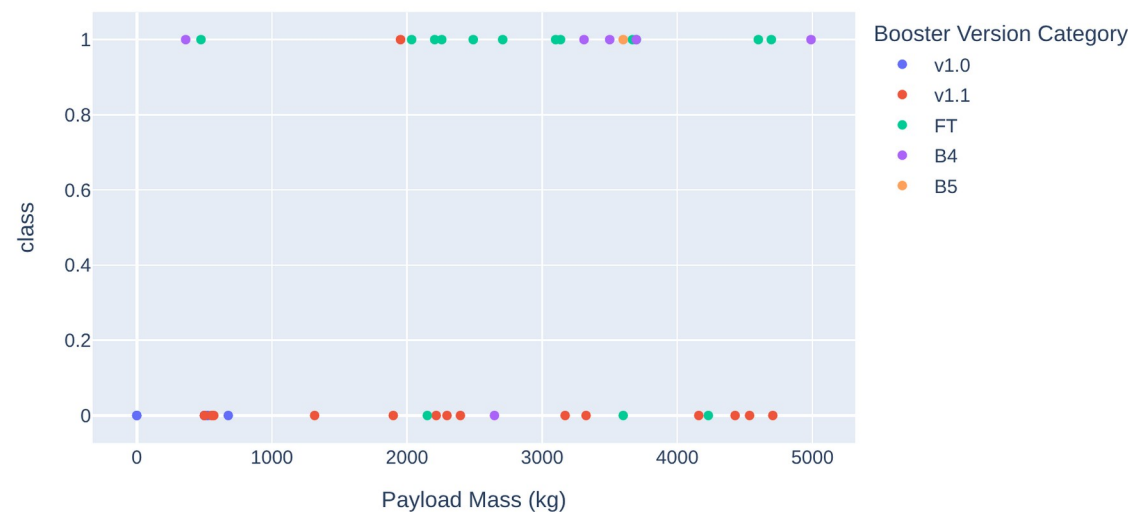
- Pie chart after selecting KSC LC-39A
- 76.9% success (10 launches)
- 23.1% failure (3 launches)
- Highest success ratio of all 4 sites; every other site is below 45%

Insight KSC LC-39A was used after the booster design matured, so its high ratio reflects timing as much as location.

Dashboard: payload vs. launch outcome for all sites

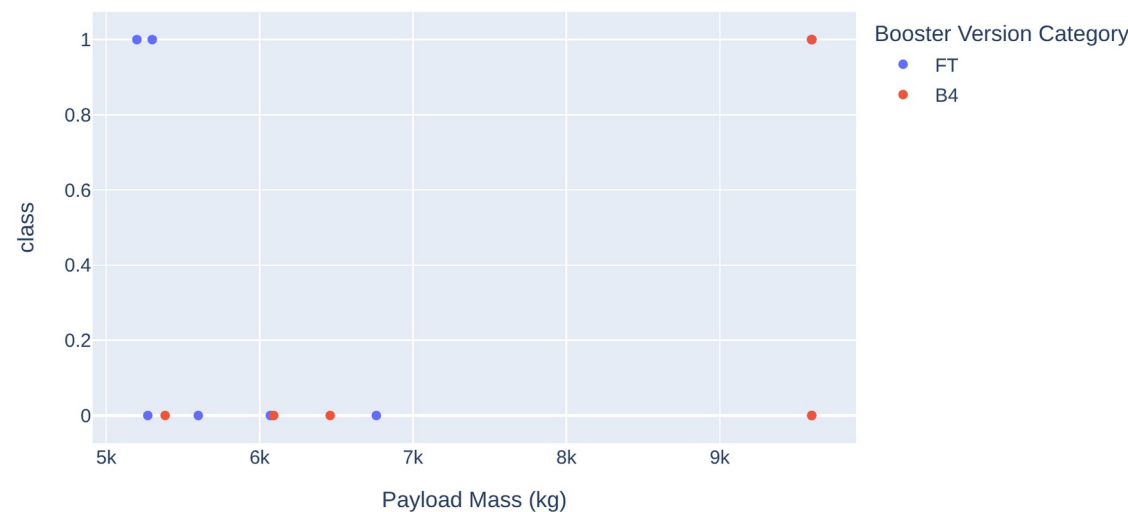
Range slider 0 - 5,000 kg

Correlation between Payload and Success for all Sites



Range slider 5,000 - 10,000 kg

Correlation between Payload and Success for all Sites

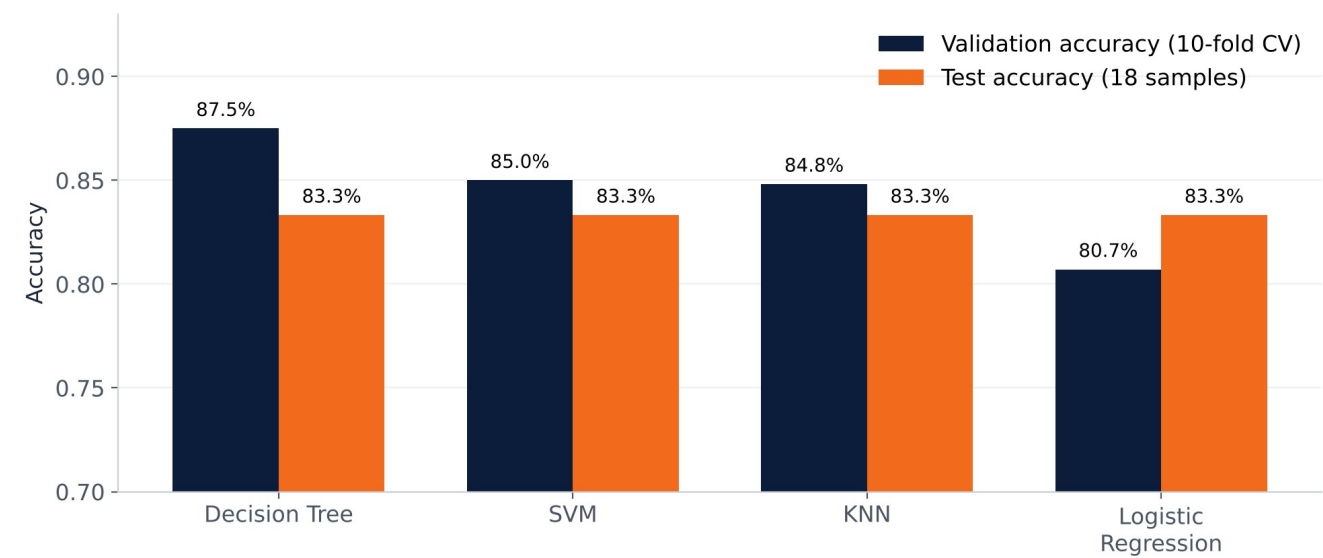


Payload (kg)	0-2,000	2,000-4,000	4,000-6,000	6,000-10,000
Success rate	23%	62%	38%	33%
Launches	13	21	13	9

- Low payloads ($\leq 5,000$ kg) succeed more often: 45% vs. 36%
- 2,000-4,000 kg has the highest success rate (62%)
- FT boosters have the best success rate (67%); v1.0 and v1.1 almost never landed

Insight Booster version matters more than payload: most low-payload failures flew on early v1.0/v1.1 boosters.

Classification accuracy



Model (best parameters)	CV accuracy	Test accuracy
Decision Tree (gini, depth 8)	87.5%	83.3%
SVM (sigmoid, C=1)	85.0%	83.3%
KNN (k=6, p=1)	84.8%	83.3%
Logistic Regression (C=0.1)	80.7%	83.3%

Which model performs best, and why

Best model: Decision Tree.

All four models reach the same test accuracy (83.3% = 15 of 18 test launches), so the test set cannot separate them.

The Decision Tree has the highest mean 10-fold cross-validation accuracy (87.5%), measured on 72 training launches in 10 folds — a more reliable estimate than 18 test samples.

It is also easy to explain with simple if/then rules.

Confusion matrix of the best model (Decision Tree)

True label	Predicted label	
	Did not land	Landed
Did not land	3 True negative	3 False positive
Landed	0 False negative	12 True positive

12 true positives • 3 true negatives

Every booster that actually landed was predicted to land: recall for 'landed' = $12/12 = 100\%$.

3 false positives • 0 false negatives

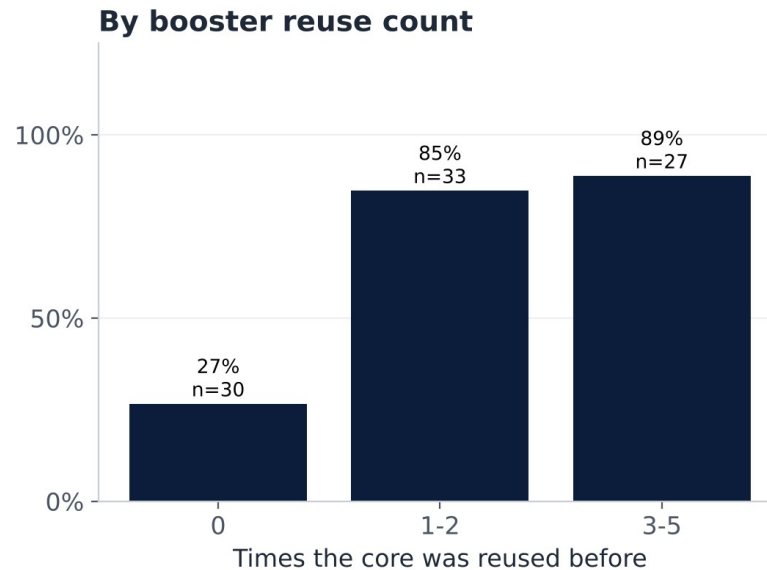
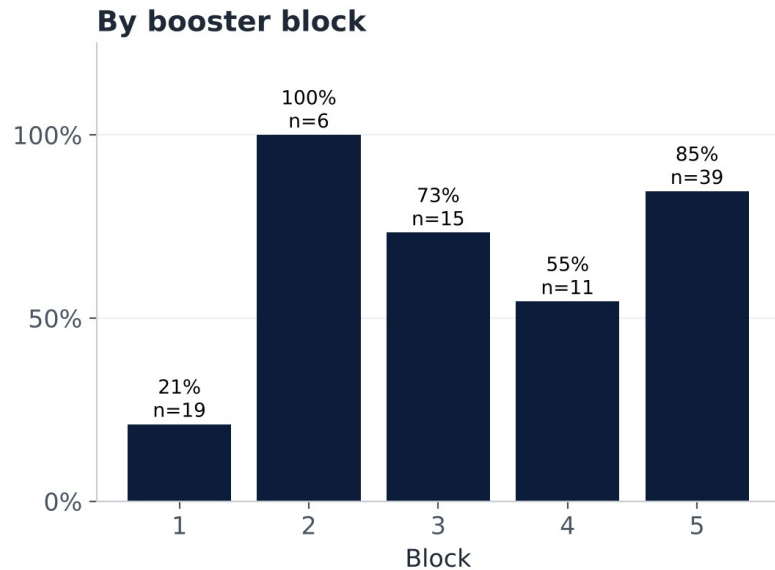
The only errors: 3 boosters that did not land were predicted to land: precision = $12/15 = 80\%$.

Explanation

The model distinguishes the classes well; its weakness is false positives (unsuccessful landings marked as successful). All four tuned models produced this same matrix.

Insight For a bidder, false positives mean underestimating SpaceX's cost risk — tuning the decision threshold is the next improvement.

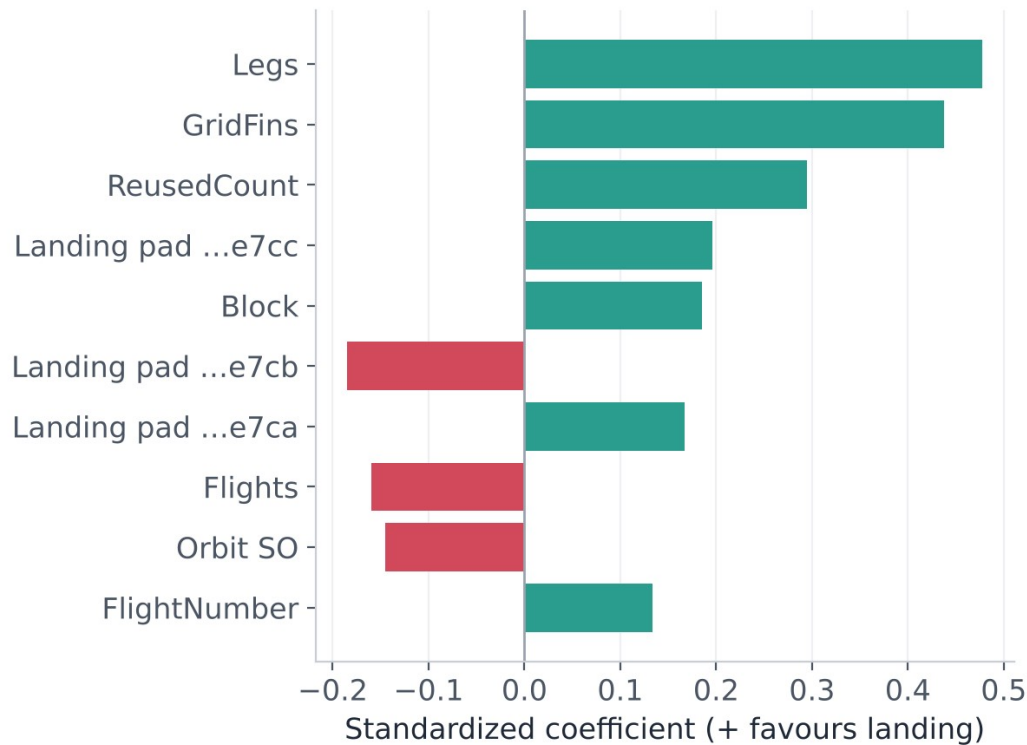
Innovative insight: reused, newer boosters land far more often



- Cores flying for the first time landed only 27% of the time (30 launches)
- Cores reused 1-2 times landed 85%; 3-5 times, 89%
- Block 5 boosters: 85% over 39 launches — the most reliable large group

Insight A booster that has landed before is very likely to land again — reuse history is a strong, cheap-to-collect predictor.

Innovative insight: landing drivers and bidding strategy



Top drivers (logistic regression)

Landing legs, grid fins and reuse count have the largest positive weights; block and flight number also help. These are hardware-readiness signals.

GTO is SpaceX's weak spot

GTO is the most frequent orbit (27 launches) but only 52% landed. When the first stage is lost, SpaceX's cost advantage shrinks.

Bidding strategy for Space Y

Target heavy GTO missions, where SpaceX is least likely to recover the booster.

Insight Serial-number dummies were excluded from the chart: they identify individual boosters and do not generalize.

Conclusion

1

Success is a learning curve

Landing rate grew from 0% (2010-13) to 90% in 2019; flight number, reuse and booster block matter most.

2

Orbit matters

ES-L1, GEO, HEO, SSO landed every time (small samples); GTO is the hardest common orbit at 52%.

3

KSC LC-39A is the best site

77% landing rate and 41.7% of all successes. All sites are coastal, close to rail and roads, and far from cities.

4

Landings are predictable

Tuned models reach 83.3% test accuracy; the Decision Tree is the best model (87.5% validation) and never misses a real landing.

5

Next steps

Add newer launches, weather and booster-history features, and tune the decision threshold to cut false positives.

Appendix

File in the GitHub repository	Content
1_spacex_data_collection_api.ipynb	SpaceX REST API collection and cleaning
2_spacex_webscraping.ipynb	Wikipedia scraping with BeautifulSoup
3_spacex_data_wrangling.ipynb	Outcome labelling (Class)
4_spacex_eda_sql.ipynb	10 SQL tasks on SQLite
5_spacex_eda_dataviz.ipynb	EDA charts and feature engineering
6_spacex_launch_site_folium.ipynb	Folium maps and distances
7_spacex_dash_app.py	Plotly Dash dashboard
8_spacex_machine_learning_prediction.ipynb	Model training and evaluation

Data notes

- Falcon 9 dataset: 90 launches, 2010-2020; model matrix 90×80 after one-hot encoding
- SQL dataset: 101 records (Spacex.csv)
- Dash / Folium dataset: 56 records
- Proximity distances come from points picked with MousePosition, so they are approximate
- Decision-tree tuning has randomness; re-runs can vary by a few points

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